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CBL Group #2 Case 1 Learning Issue 1: Tetanus and Tetanus Vaccine Status Importance

Learning Issue

What is tetanus? Why is it important to know the tetanus vaccine status of patients who present with open wounds in the emergency room?

Main Points

Tetanus is a bacterial infection caused by the bacterium *Clostridium tetani*. Tetanus can cause symptoms such as trismus, muscle spasms, and muscle rigidity. In the emergency room, it is critical to know a patient's tetanus vaccine status to prevent a potentially fatal *Clostridium tetani* infection.

Key Terms

Tetanus - A bacterial infection that affects the nervous system, causing muscle spasms and muscle rigidity. The infection is not contagious.

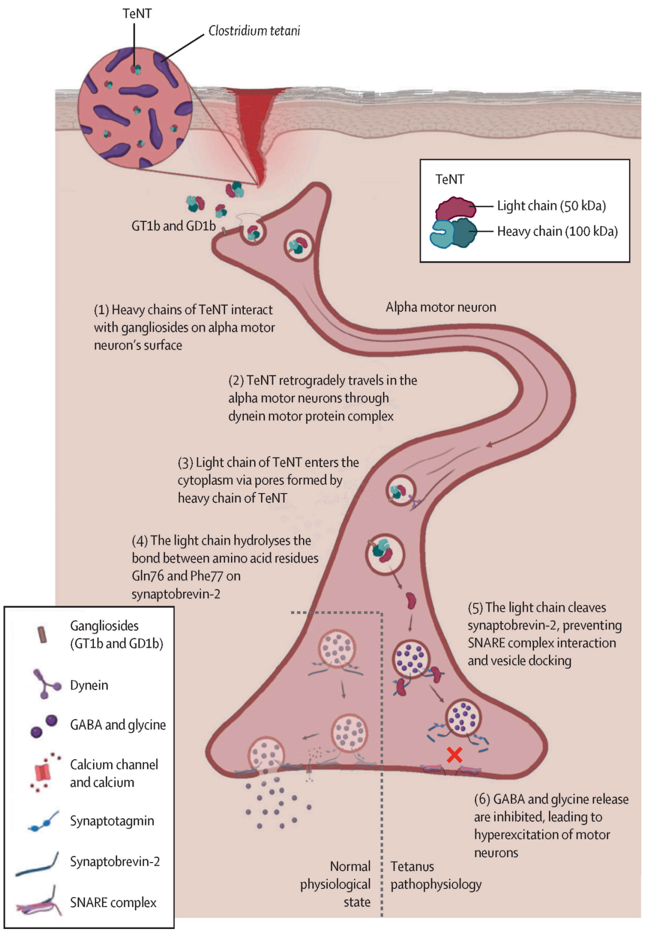
Trismus - Tightening of jaw muscles that affects the ability to open one's mouth.

Clostridium tetani - A bacterium that causes tetanus by releasing tetanospasmin neurotoxin.

Relevant Research

Tetanus is a bacterial infection caused by the neurotoxin tetanospasmin produced by the bacterium *Clostridium tetani*. The symptoms of tetanus in adults include trismus, muscle spasms, and muscle stiffness [1]. Muscle spasms and muscle stiffness can also lead to respiratory symptoms such as difficulty breathing or even respiratory failure [1]. Tetanus is acquired through a contaminated open wound. Common contaminants include dirt, dust, feces, and saliva [2]. If deep and superficial wounds are not properly cleaned or irrigated with saline, *Clostridium tetani* spores germinate into vegetative rod-shaped bacteria that release the tetanospasmin neurotoxin [1].

The tetanospasmin neurotoxin has two subunits: a heavy chain and a light chain. The neurotoxin travels to the ventral horn of the spinal cord, where the heavy chain will interact with the glycocalyx of neurons, which allows the neurotoxin to attach to the cells [1]. The heavy chain will then create a channel in the neuron cell membrane to allow the light chain to cross into the cytoplasm [1]. In the cytoplasm, the light chain cleaves neuronal SNARE proteins, which function to fuse synaptic vesicles to the plasma membrane [3]. Without the SNARE complex, neuronal signaling is lost, which results in uncontrolled hyperactivity of motor neurons [1].



Tetanus pathophysiology (Figure 3 of [1])

Generalized tetanus most often presents as muscle spasms and muscle rigidity. These symptoms usually start as trismus and neck stiffness, but frequently progress to more severe spasms and respiratory symptoms, which can be life-threatening [1].

It is important to know the tetanus vaccine status in patients, especially those who present with open wounds, because tetanus is a noncommunicable disease that is contracted from environmental contamination [4]. Since herd immunity cannot contribute to the prevention of tetanus, the tetanus vaccine is crucial for individual protection [4].

Clinical Relevance

Our patient has an open wound from a motor vehicle crash, which is a risk factor for tetanus.

Furthermore, his tetanus vaccine status is unknown. Because of his open wound and unknown last tetanus vaccine, I would recommend administration of the tetanus vaccine. If the wound is dirty or contaminated, I would also recommend administration of tetanus immune globulin (TIG) [5].

References

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[2] World Health Organization. (2024, February 12). *Tetanus: Immunization*. https://www.who.int/news-room/questions-and-answers/item/tetanus-immunization

[3] Rodrigo, C., Fernando, D., & Rajapakse, S. (2014). Pharmacological management of tetanus: An evidence-based review. *Critical Care, 18*(2), 217. https://doi.org/10.1186/cc13797

[4] Centers for Disease Control and Prevention. (2024, October 22). *Chapter 16: Tetanus*. *Manual for the Surveillance of Vaccine-Preventable Diseases*. https://www.cdc.gov/surv-manual/php/table-of-contents/chapter-16-tetanus.html

[5] Centers for Disease Control and Prevention. (2025, June 10). *Clinical guidance for wound management to prevent tetanus*. https://www.cdc.gov/tetanus/hcp/clinical-guidance/index.html